

Rethinking case attraction on Ancient Greek infinitive clauses

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FFLCH-USP

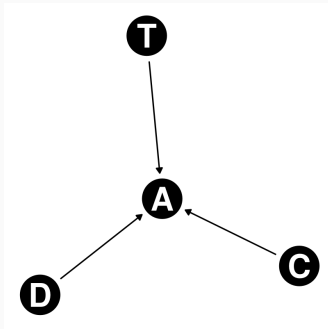


- (1) a. συμβουλεύει **τῷ Ξενοφῶντι ἐλθόντα** εἰς Δελφοὺς
ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
He advises Xenophon to go to Delphi and ask the god
about the travel. (Xen. *Anab.* 3.1.5)
- b. ἀφῆκε **μοι ἐλθόντι** πρὸς ὑμᾶς λέγειν τάληθῃ.
He allowed me to go and speak the truth in front of you.
(Xen. *Hell.* 6.1.13)

Direct Acyclic Graph: Agreement

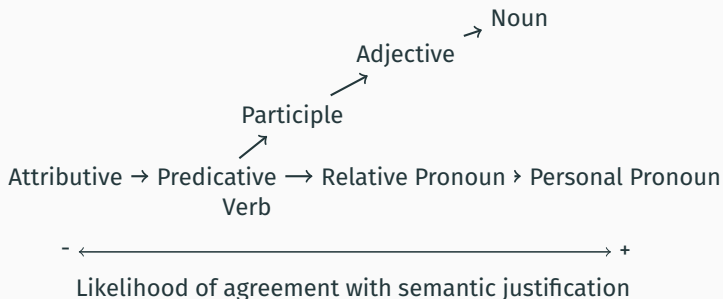
Framework: Corbett (2006)

Features of the **C**ontroller, **T**arget and **D**omain “cause”
non-canonical **A**greement

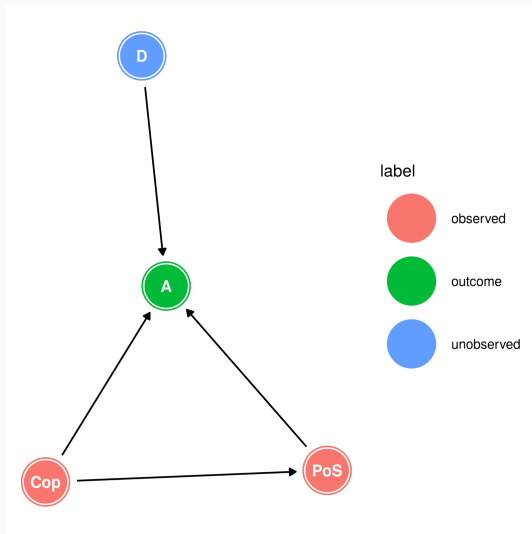


Domain of attraction: Part of speech and copular infinitive

(2) Agreement and Predicate Hierarchies (Corbett 2006, p. 233):

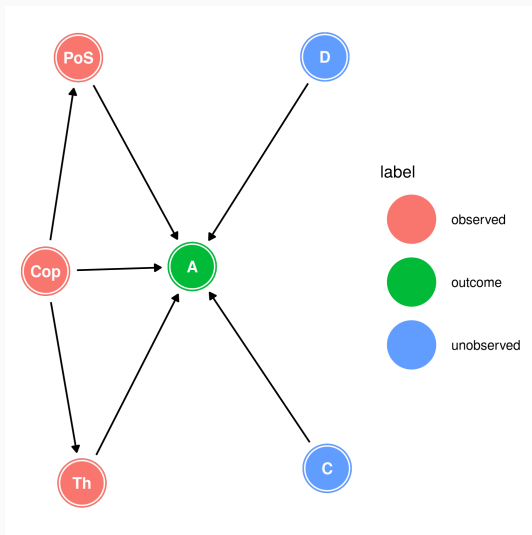


The domain of attraction: DAG

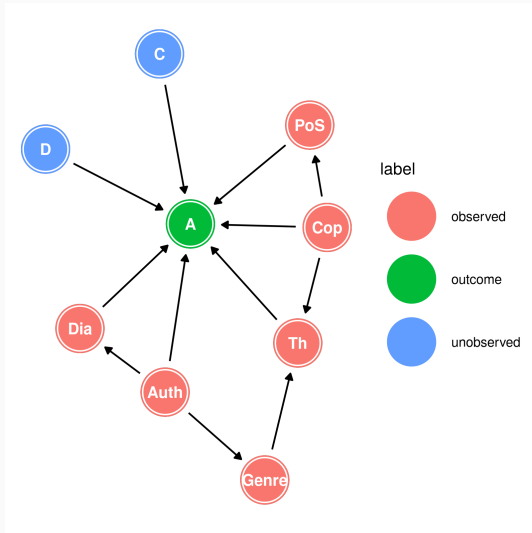


- (3) ἀλλ' ἐξαρκῇ **σοι** τὰ ἔργα αὐτῶν **ὁρῶντι** σέβεσθαι καὶ τιμᾶν τοὺς θεοὺς.
But it is enough for you to look at these deeds and respect the gods. (Xen. *Mem.* 4.3.13)

The controller of attraction: DAG



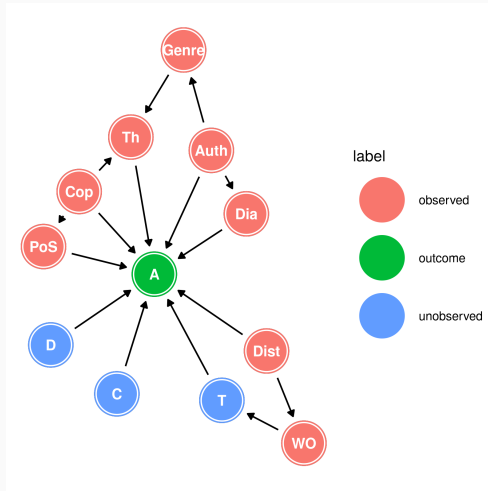
Issues of dialect and authorship: DAG



The target of attraction: distance and precedence

- (4) ὥς **ἀγαθοῖς** τε **ὕμῃν** προσήκει εἶναι.
That you may be good people. (Xen. *Anab.* 3.2.11)

The target of attraction: final DAG



Generalized linear model

$$A \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha + \beta_{\text{Cop}} \text{Cop}_i + \beta_{\text{Dist}} \text{Dist}_i + \beta_{\text{Wo}} \text{Prec}_i$$

$$+ \beta_{\text{PoS}} \sum_{j=0}^{\text{PoS}_i-1} \delta_{j\text{PoS}} + \beta_{\text{Th}} \sum_{j=0}^{\text{Th}_i-1} \delta_{j\text{Th}}$$

$$+ \beta_{\text{Auth}_i} + \beta_{\text{Dia}_i}$$

$$\alpha \sim \text{Normal}(0, 1)$$

$$\beta_{\text{Cop}}, \beta_{\text{Dist}}, \beta_{\text{Prec}} \sim \text{Normal}(0, 1)$$

$$\beta_{\text{PoS}}, \beta_{\text{Th}} \sim \text{Normal}(0, 0.5)$$

$$\delta_{\text{PoS}}, \delta_{\text{Th}} \sim \text{Dirichlet}(2)$$

$$\beta_{\text{Auth}} = \mathbf{Z}_{\text{Auth}} * \sigma_{\text{Auth}}$$

$$\beta_{\text{Dia}} = \mathbf{Z}_{\text{Dia}} * \sigma_{\text{Dia}}$$

$$\mathbf{Z}_{\text{Auth}}, \mathbf{Z}_{\text{Dia}} \sim \text{Normal}(0, 1)$$

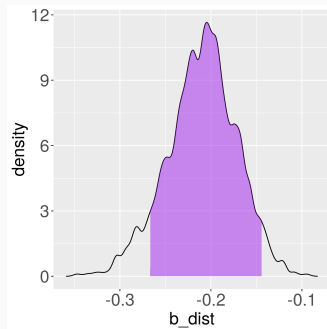
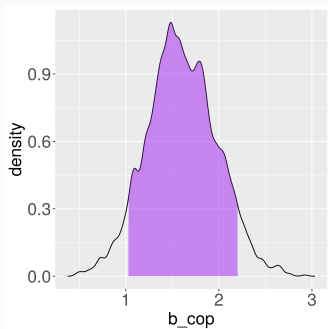
$$\sigma_{\text{Auth}}, \sigma_{\text{Dia}} \sim \text{Exponential}(1)$$

A word on Ordered Categorical Predictors

PoS	Cummulative effect
Participle	β_{PoS}
Adjective	$\beta_{\text{PoS}} + \delta_{\text{Adj}}$
Noun	$\beta_{\text{PoS}} + \delta_{\text{Adj}} + \delta_{\text{Noun}}$

Copula and Distance

Effect	mean	sd	5.5%	94.5%	rhat
β_{Copula}	1.60	0.37	1.02	2.20	1
β_{Dist}	-0.21	0.04	-0.28	-0.15	1



Part of Speech and the Predicate Hierarchy

Effect	mean	sd	5.5%	94.5%	rhat
β_{PoS}	-0.01	0.35	-0.59	0.52	1
δ_{Adj}	0.50	0.23	0.12	0.88	1
δ_{Noun}	0.50	0.23	0.12	0.88	1

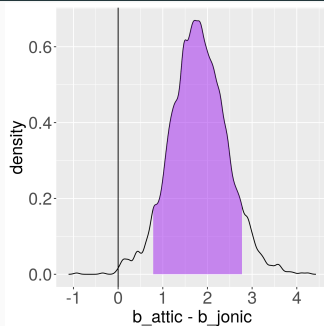
PoS	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Participle	β_{PoS}	-0.01	0.35	-0.59	0.52	1
Adjective	$\beta_{\text{PoS}} + \delta_{\text{Adj}}$	0.49	0.45	-0.22	1.19	1
Noun	$\beta_{\text{PoS}} + \delta_{\text{Adj}} + \delta_{\text{Noun}}$	0.99	0.35	0.41	1.52	1

Thematic Role of the Controller

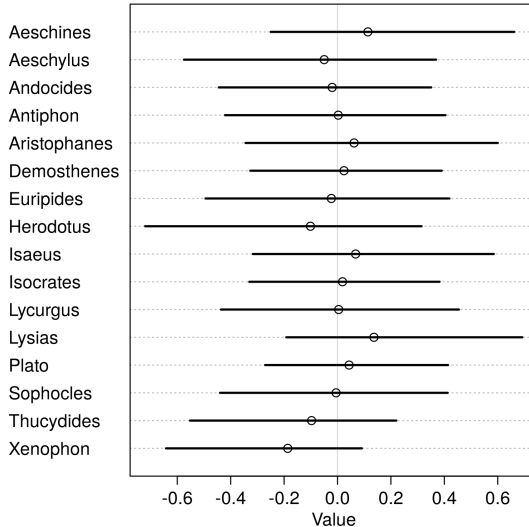
Effect	mean	sd	5.5%	94.5%	rhat
β_{Theta}	1.64	0.30	1.17	2.12	1
δ_{Exp}	0.29	0.14	0.09	0.52	1
δ_{Agent}	0.71	0.14	0.48	0.91	1

Theta	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Recipient	β_{Rec}	1.64	0.30	1.17	2.12	1
Experiencer	$\beta_{\text{Rec}} + \delta_{\text{Exp}}$	1.92	0.33	1.41	2.44	1
Agent	$\beta_{\text{Rec}} + \delta_{\text{Exp}} + \delta_{\text{Agent}}$	2.64	0.30	2.17	3.12	1

Effect	mean	sd	5.5%	94.5%
$\bar{\beta}_{\text{Dia}}$	-0.22	0.86	-1.58	1.15
σ_{Dia}	1.36	0.78	0.50	2.78
β_{Attic}	0.45	0.87	-0.93	1.80
β_{Jonic}	-1.35	0.97	-2.94	0.18
$\beta_{\text{Attic}} - \beta_{\text{Jonic}}$	1.80	0.63	0.78	2.78



Authorship



Thank you!

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Data and code available at

`github.com/caiogeraldes/icag12025`